



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

23-08-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PART-1 : PHYSICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	C,D	C	B,C,D	A,C,D	A,B,C	A,C	A,C	A,C,D	C	B
	Q.	11	12								
	A.	D	A								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.00	1.50	84.80 to 84.90	4.50	336.00	70.60 to 70.70				

PART-2 : CHEMISTRY

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B,C	B,D	C,D	A,D	A,C	C,D	A,D	A,C	C	D
	Q.	11	12								
	A.	B	A								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.00	0.12 or 0.13	1.00	8.00	2.00	2.00				

PART-3 : MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	A,B,C,D	A,D	A,C,D	B,C	A,C	A,B	A	A,B,C,D	A	A
	Q.	11	12								
	A.	C	D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	9.00	4.00	4.00	5.94	2.41	2.00				



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PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B,C)

2. Ans.(B,D)

3. Ans.(C,D)

4. Ans.(A,D)

5. Ans.(A,C)

6. Ans. (C,D)

7. Ans.(A, D)

8. Ans. (A,C)

(A) It is always 10 for any solution

$$(B) \frac{\%w/w}{\%w/w} = \frac{1}{d} = 0.8 \Rightarrow d = 1.25 \text{ gm/ml}$$

(C) 1000 ml solution contains 2 mol solute
1250 gm solution contains 50 gm solute

$$W_{\text{solvent}} = 1200 \text{ gm}$$

$$\text{Molarity} = \frac{2}{1200} \times 1000 = \frac{10}{6}$$

9. Ans.(C)

10. Ans.(D)

11. Ans. (B)

12. Ans. (A)

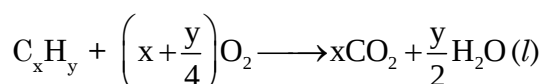
SECTION-II

1. Ans.(3.00)

2. Ans.(0.12 or 0.13)

3. Ans.(1.00)

4. Ans.(8.00)



$$\begin{array}{c} a \\ \downarrow \\ 0 \end{array} \quad V$$

$$V - a(x + y/4) \quad ax$$

$$\text{vol contraction} = V_i - V_f$$

$$= (a + V) - \left[V - a\left(x + \frac{y}{4}\right) + ax\right]$$

$$= a + a\frac{y}{4}$$

Given that

$$\text{volume contraction} = 2.5 V_{H.C}$$

$$a + a\frac{y}{4} = 2.5a$$

$$y = 6$$

Given that after passing KOH

$$\text{Vol contraction} = 2.0 V_{H.C}$$

$$ax = 2a$$

$$x = 2$$



$$\text{no. of atom } 2 + 6 = 8$$

5. Ans.(2.00)

6. Ans.(2.00)

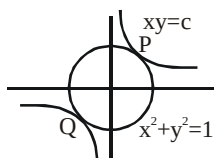
PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,B,C,D)

Sol.



To find point of contact solve both the curves.

$$x^2 + \frac{c^2}{x^2} = 1 \Rightarrow x^4 - x^2 + c^2 = 0$$

$$\text{both touch each other} \Rightarrow D = 0$$

$$1 - 4c^2 = 0 \Rightarrow c = \frac{1}{2} \quad (c > 0)$$

$$\Rightarrow \left(x^2 - \frac{1}{2}\right)^2 = 0 \Rightarrow x = \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$$

$$P\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), Q\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

$$\Rightarrow PQ = 2$$

$$\text{Slope of tangent} \Rightarrow -\frac{(1/\sqrt{2})}{(1/\sqrt{2})} = -1$$

Length of latus rectum of $xy = c$ is 2

2. Ans. (A,D)

$$A(2,0)$$

Sol.
$$\frac{B(0,2)}{2x + 3y + 1 = 0}$$

|PA - PB| is minimum

$\Rightarrow PA = PB$

intersection point of $2x + 3y + 1 = 0$

& $y - 1 = x - 1$

$$\Rightarrow \left(-\frac{1}{5}, -\frac{1}{5}\right)$$

|PA - PB| is maximum

$\Rightarrow$ intersection point of

$2x + 3y + 1 = 0$ & $x + y = 2$

$\Rightarrow (7, -5)$

3. Ans. (A,C,D)

Sol. For Q and 'R' $t_1 t_2 = 2$ and for S $t_3 = -(t_1 + t_2)$ and $P(at_1 t_2, a(t_1 + t_2))$

$\therefore$ If $Q(1,2)$, then $R(4,4)$, $S(9,-6)$ and $P(2,3)$ quadrilateral PQRS is cyclic and PS will be diameter of circle

$\Rightarrow$ circumcentre of ΔPQR and ΔQRS is

$$\left(\frac{11}{2}, -\frac{3}{2}\right)$$

4. Ans. (B,C)

Sol. Any tangent to the ellipse is

$$y = mx \pm \sqrt{24m^2 + 16}$$

If it passes through (h, k) then

$$(k - mh)^2 = 24m^2 + 16$$

$$\Rightarrow m^2(h^2 - 24) - 2m hk + k^2 - 16 = 0$$

$$\therefore m_1 m_2 = \frac{k^2 - 16}{h^2 - 24} = 1$$

$$\Rightarrow \text{Locus of } P(h,k) \text{ is } \boxed{x^2 - y^2 = 8}$$

Now check Options

5. Ans. (A,C)

Sol. $I + F = (15 + \sqrt{224})^n = {}^nC_0(15)^n$

$$+ {}^nC_1(15)^{n-1}(\sqrt{224})^1 + \dots + {}^nC_n(15)^0(\sqrt{224})^n$$

$$\text{Let } F' = (15 - \sqrt{224})^n$$

$$= {}^nC_0(15)^n - {}^nC_1(15)^{n-1}(\sqrt{224})^1 + \dots$$

$$\Rightarrow I + F + F' = \text{even}$$

$$\Rightarrow 0 < F + F' < 2 \Rightarrow F + F' = 1$$

$$\Rightarrow F' = 1 - F$$

$$\Rightarrow I = \text{odd and } x(1 - F)$$

$$= (15 + \sqrt{224})^n (15 - \sqrt{224})^n = 1$$

6. Ans. (A,B)

Sol.
$$\frac{n+1}{1 + \left|\frac{x}{y}\right|} = \frac{19+1}{1 + \left|\frac{2}{3x}\right|}, x = 1$$

$$= \frac{20 \times 3}{5} = 12$$

$\Rightarrow T_{12}$ & T_{13} are numerically greatest.

7. Ans. (A)

Sol. required sum is coefficient of x^{10} in the expansion of $(1+x)^{30}(1+x)^{20}(1+x)^{10}$ i.e. ${}^{60}C_{10}$

8. Ans. (A,B,C,D)

Sol. $4\ell^2 = 2b^2 + 2c^2 - a^2 = b^2 + c^2 + (b^2 + c^2 - a^2)$

$$= b^2 + c^2 + 2bc \cos A$$

$$= b^2 + c^2 - 2bc \cos A + 4bc \cos A$$

$$= a^2 + 4bc \cos A = a^2 + 4b(c \cos A)$$

$$= a^2 + 4b(b - a \cos C)$$

$$= a^2 + 4b^2 - 4ab \cos C$$

$$= (a + 2b)^2 - 4ab - 4ab \cos C$$

$$= (a + 2b)^2 - 8ab \cos^2 \frac{C}{2}$$

$$= (a + 2b)^2 - 8ab \frac{s(s-c)}{ab}$$

$$= (a + 2b)^2 - 8s(s-c)$$

Solution Q.9 and Q.10

Sol. Both roots will be real.

$$\text{If } \frac{-b}{2a} > K$$

$\Rightarrow$ one root must be greater than K.

Now, using graph one can find location of other root.

9. Ans. (A)

10. Ans. (A)

Solution Q.11 and Q.12

Sol. (I) The equation has a meaning if $2x \neq n\pi$

$$3x \neq n\pi. 2x \neq (2n+1)\frac{\pi}{2}$$

For these values, the equation can be written as.

$$3\cot 2x - 3\cot 3x = \tan 2x + \cot 2x$$

$$\Rightarrow 3\left(\frac{\cos 2x}{\sin 2x} - \frac{\cos 3x}{\sin 3x}\right) = \frac{\sin 2x}{\cos 2x} + \frac{\cos 3x}{\sin 2x}$$

$$\Rightarrow \frac{3\sin x}{\sin 2x \sin 3x} - \frac{1}{\sin 2x \cos 2x}$$

$$\Rightarrow 3\sin x \cos 2x = \sin 3x$$

$$\Rightarrow \sin x(3 - 4\sin^2 x - 3\cos 2x) = 0$$

$$\Rightarrow \sin x \sin^2 x = 0$$

$$\Rightarrow \sin x = 0 \Rightarrow x = n\pi$$

But for $x = n\pi$, the equation has no meaning

$\Rightarrow$ The given equation has no solution

$$(II) \sin^{10} x + \cos^{10} x = \frac{29}{16} \cos^4 2x$$

$$\left(\frac{1 - \cos 2x}{2}\right)^5 + \left(\frac{1 + \cos 2x}{2}\right)^5 = \frac{29}{16} \cos^4 2x$$

$$= \frac{29}{16} \cos^4 2x$$

$$\Rightarrow \frac{1}{32} \times 2\{1 + 10\cos^2 2x + 5\cos^4 2x\}$$

$$\Rightarrow (2\cos^2 2x - 1)(12\cos^2 2x + 1) = 0$$

$$\Rightarrow \cos^2 2x = \frac{1}{2} \Rightarrow x = \frac{n\pi}{4} + \frac{\pi}{8}$$

$$(III) \text{ Let } \sin 2x = y, \text{ then } 1 + y^4 = 17(1 + y)^4$$

$$\Rightarrow \frac{1}{y^2} + y^2 = \frac{17}{y^2} (1 + 2y + y^2)^2$$

$$= 17 \left(\frac{1}{y} + 2 + y \right)^2$$

$$\Rightarrow 8z^2 + 34z + 35 = 0, \text{ where } z = y + \frac{1}{y}$$

$$\Rightarrow \sin 2x = -\frac{1}{2} \text{ is the only possibility}$$

$$\Rightarrow x = 105^\circ, 165^\circ, 285^\circ, 345^\circ$$

(IV) Given equation is

$$\cos(2x - 59^\circ) = \sin(x + 20^\circ)$$

$$\Rightarrow \cos(2x - 59^\circ) = \cos(70^\circ - x)$$

$$\Rightarrow 2x - 59^\circ = (2n)180^\circ \pm (70^\circ - x)$$

$$2x - 59^\circ = (2n)180^\circ + (70^\circ - x) \text{ or}$$

$$2x - 59^\circ = (2n)180^\circ - (70^\circ - x)$$

$$\Rightarrow x = \frac{(2n)180^\circ}{3} + 43^\circ, n \in \mathbb{Z} \text{ or}$$

$$x = (2n)180^\circ - 11^\circ, n \in \mathbb{Z}$$

11. Ans. (C)

12. Ans. (D)

SECTION-II

1. Ans. 9.00

$$\text{Sol. } p_1 p_2 = b^2 = 9$$

2. Ans. 4.00

$$\text{Sol. } C_1 \rightarrow aC_1, C_2 \rightarrow bC_2 \text{ \& } C_3 \rightarrow cC_3$$

$$\frac{1}{abc} \begin{vmatrix} a(b^2 + c^2) & ab^2 & ac^2 \\ a^2b & b(c^2 + a^2) & bc^2 \\ a^2c & b^2c & c(a^2 + b^2) \end{vmatrix}$$

$$= \begin{vmatrix} b^2 + c^2 & b^2 & c^2 \\ a^2 & c^2 + a^2 & c^2 \\ a^2 & b^2 & a^2 + b^2 \end{vmatrix}$$

Use $C_1 > (C_1 + C_2 + C_3)$ and solve

3. Ans. 4.00

Sol. Let equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ it passes through $(0, 0) \Rightarrow C = 0$

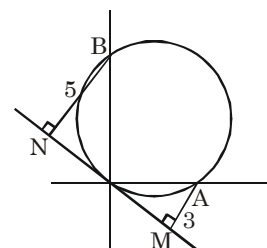
tangent at origin = $gx + fy = 0$

$A(-2g, 0), B(0, -2f)$ on x-axis & y-axis

$$AM = \left| \frac{-2g^2}{\sqrt{g^2 + f^2}} \right| = 3$$

$$BN = \left| \frac{-2f^2}{\sqrt{g^2 + f^2}} \right| = 5$$

$$\Rightarrow \sqrt{g^2 + f^2} = 4$$



4. Ans. 5.94

Sol. $t_r = \frac{2r+1}{1^2+2^2+\dots+r^2}$

$$= \frac{6}{r(r+1)} = 6\left(\frac{1}{r} - \frac{1}{r+1}\right)$$

$$\Rightarrow S_n = 6\left(1 - \frac{1}{n+1}\right) = \frac{6n}{n+1}$$

5. Ans. 2.41

Sol. Let $S = \sum_{n=1}^8 \sqrt{3+\sqrt{n}}$, $T = \sum_{n=1}^8 \sqrt{3-\sqrt{n}}$.

Note that $\sqrt{a+b+2\sqrt{ab}} = \sqrt{a} + \sqrt{b}$ for positive real numbers a and b . In particular, for each fixed positive integer n with $1 \leq n \leq 8$, choose $a > b > 0$ satisfying $a+b=6$, $ab=n$, then a, b are roots of the equations $x^2 - 6x + n = 0$ i.e.

$$a = 3 + \sqrt{9-n}, \quad b = 3 - \sqrt{9-n}$$

$$\text{so } \sqrt{2}S = \sum_{n=1}^8 \sqrt{6+2\sqrt{n}}$$

$$= \sum_{n=1}^8 \left(\sqrt{3+\sqrt{9-n}} + \sqrt{3-\sqrt{9-n}} \right)$$

$$= \sum_{n=1}^8 \left(\sqrt{3+\sqrt{n}} + \sqrt{3-\sqrt{n}} \right) = S + T$$

$$\text{Thus, } (\sqrt{2}-1)S = T \Rightarrow \frac{S}{T} = \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}$$

6. Ans. 2.00

Sol. As we know that $\cot\theta - \tan\theta = 2\cot 2\theta$

$$\cot\theta - \tan\theta = 2\cot 2\theta$$

$$\cot\theta + \tan\theta = 2\operatorname{cosec} 2\theta$$

$$\frac{(\cot\theta - \tan\theta - 2\operatorname{cosec} 2\theta)(\cot\theta + \tan\theta)}{(2\cot 2\theta - \cot\theta - \tan\theta)\operatorname{cosec} 2\theta}$$

$$= \frac{2(\cot 2\theta - \operatorname{cosec} 2\theta) \cdot 2\operatorname{cosec} 2\theta}{(2\cot 2\theta - 2\operatorname{cosec} 2\theta)\operatorname{cosec} 2\theta} = 2$$